

Dynamic Reduction Network for Resource-Efficient RGBT Tracking

IEEE Publication Technology, *Staff*, *IEEE*,

Abstract—RGBT tracking aims to leverage the complementary information from RGB and thermal infrared (TIR) modalities, offering robust tracking in diverse environments. However, most existing methods only focus on how to utilize the complementary information of multiple modalities to improve accuracy, while ignoring the computational overhead brought by multiple modalities. Furthermore, in certain scenarios, low quality modal information may impair the performance of the tracker. To remedy this situation, a novel Dynamic RGBT Tracking Network (DRNet) is presented to trade off between a resource-efficient RGBT tracker and robust tracking. In particular, we design a lightweight modality exit module to perform dynamic exit of input modalities, thus effectively reducing unnecessary computation of low-quality modalities. To enhance the robustness of modality exit module, we incorporate a layer of shared Vision Transformer (ViT) block within the module for preliminary feature extraction. Additionally, we adopt the layer reuse technique, where the intermediate layer outputs of the modality exit module are utilized for subsequent feature extraction in the backbone network, thereby avoiding redundant computations. Lastly, to stabilize the training process of the modality exit module, we devise a pseudo-label training strategy, where pseudo-labels for modality usage are generated beforehand for training purposes. Extensive experiments on three RGBT tracking benchmarks demonstrate the effectiveness of DRNet, which achieves performance comparable to or even better than existing algorithms while significantly improving efficiency. Compared to method with similar performance, DRNet reduces the number of parameters by 54.4% and achieves an inference speed that is up to 2.2 times faster. We will release the code on xxx.

Index Terms—Computer Vision, RGBT Tracking, Dynamic Network, Early Exit.

I. INTRODUCTION

VISUAL object tracking task aims to estimate the target position and size in subsequent frames using the target state from the initial frame. In recent years, a large number of excellent works [1]–[5] have emerged. However, the RGB modality is susceptible to variations in lighting conditions. For instance, extreme lighting conditions such as nighttime and strong sunlight, as well as adverse weather conditions like rain and snow, can cause the images captured in the RGB modality to lose discriminative information, ultimately leading to tracking failure. Due to the insensitivity of thermal infrared imaging to lighting conditions, RGBT tracking, which leverages the complementary information from both RGB and thermal infrared modalities, has achieved reliable target tracking in all-weather conditions and has garnered increasing attention [6]–[14].

Currently, RGBT tracking models are primarily categorized into two types: those based on Convolutional Neu-

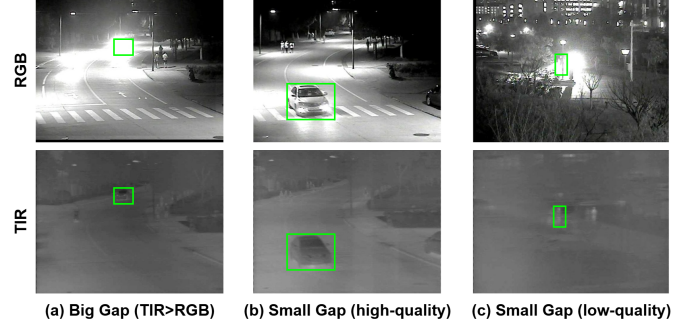


Fig. 1. Comparison of modal quality in different scenarios.

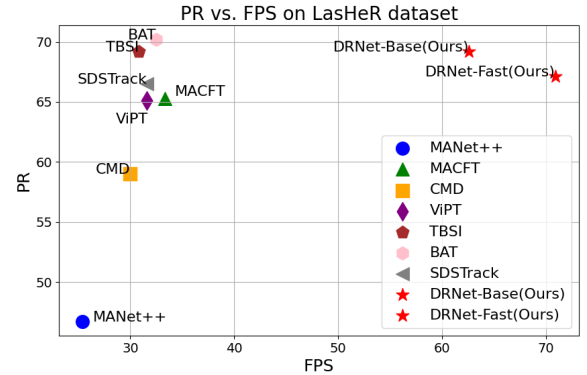


Fig. 2. PR vs. FPS on LasHeR [15] dataset. Our method significantly enhances inference speed and conserves computational resources while maintaining tracking accuracy.

ral Networks (CNNs) [16]–[19] and those based on Vision Transformers (ViTs) [13], [14], [20], [21]. Methods based on CNNs often necessitate the design of complex network structures to compensate for the difficulty of CNNs in capturing global features. For example, MANet++ [17] designs a multi-adapter network architecture that utilizes a Generality Adapter to extract modality-shared information and a Modality Adapter to extract modality-specific information. Subsequently, it employs a late fusion strategy to integrate these two types of information, enabling robust multi-modal object tracking. Methods based on ViTs leverage the global information extraction capability of ViT [22] block to easily focus on global information, and they are capable of joint feature extraction and relationship modeling. Therefore, their model structures are relatively simple. For instance, TBSI [14] proposes a Template-Bridged Search region Interaction module that utilizes a cross-attention mechanism to achieve modal interaction, with the fused template features serving as the medium. BAT [13] proposes a bi-directional adapter for prompt fusion of multi-modal features, thereby achieving

This paper was produced by the IEEE Publication Technology Group. They are in Piscataway, NJ.

Manuscript received April 19, 2021; revised August 16, 2021.

robust object tracking. SDSTrack [21] employs lightweight adapters to transfer the feature extraction capability of network trained on RGB images to other modalities, and designs a complementary masked patch distillation strategy to achieve robust tracking.

Despite their remarkable performance in tracking accuracy, traditional RGBT tracking methods often suffer from issues of low resource utilization and inefficient inference due to redundant processing of multi-modal data. This is because existing methods typically treat all modalities equally without considering the varying reliability and importance of each modality in different scenarios. In fact, due to the diversity of tracking scenarios, not all modalities are always necessary. We present three representative scenarios in Fig. 1:

i) **Modality interference:** As depicted in Fig. 1(a), in certain scenarios, some modalities may be unreliable and useless, and traditional multi-modal fusion approaches can result in a waste of computational resources and may even degrade the information of high-quality modalities.

ii) **Modality redundancy:** As shown in Fig. 1(b), when both modalities are reliable, perhaps a single modality is sufficient for the network to easily locate the target.

iii) **Modality unreliable:** In Fig. 1(c), both modalities exhibit relatively low quality. Therefore, the complementary information from the two modalities can be utilized to identify and locate the target. Alternatively, given that neither modality is reliable, it is also a viable option to rely solely on the information from either modality to reduce computational load, potentially without compromising tracking performance.

To address the aforementioned issues, we propose a novel Dynamic Reduction Network (DRNet) for resource-efficient RGBT tracking. Different from traditional RGBT tracking methods that treat all modality inputs equally, DRNet can achieve dynamic forward propagation based on current input conditions. Specifically, the Modality Exit (ME) module we proposed can adaptively determine the modality usage, i.e., whether to exit a particular modality, based on the quality of the input from the current modality. During the design of the module, to enhance the robustness of the ME module in discriminating the quality of input images, we integrated a ViT block within the module for preliminary feature extraction. Furthermore, to mitigate the computational overhead introduced by the additional ViT block, we employed layer reuse technology, enabling the output of the ViT layer within the module to be utilized for subsequent feature extraction in the backbone network. Lastly, to stabilize the training process of the ME module and enhance its training effectiveness, we devised a pseudo-label training strategy, utilizing pre-generated modality-usage pseudo-labels to train the ME module.

Our main contributions can be summarized as follows:

- We propose a novel dynamic reduction RGBT tracking network, which employs a modality exit module to adaptively discard unreliable interfering modality as well as redundant modality.
- A pseudo-label training approach is introduced to stabilize the training process of the modality exit module. Additionally, we introduce a layer reuse technique to

enhance the robustness of modality exit module while reducing the wastage of computational resources.

- To the best of our knowledge, this is for the first time that early exit technology is introduced in RGBT tracking tasks. Extensive experiments show that our method preserves high tracking accuracy while conserving computational resources and substantially accelerating inference speed.

II. RELATED WORK

A. RGBT Tracking

The research focus of RGBT object tracking lies in leveraging the complementary information from the two modalities to achieve robust target tracking. To this end, numerous excellent works [6], [10], [13], [14], [16], [20], [23]–[25] have been proposed. For instance, APFNet [18] designs an attribute-based progressive fusion network that extracts attribute-specific fusion features and performs adaptive aggregation. Subsequently, an enhanced fusion transformer is utilized to augment both the aggregated features and modal-specific features, thereby bolstering the robustness of the tracker against challenging attributes. QAT [6] learns to generate modal pseudo reliability labels in a supervised manner, and then employs these generated pseudo labels to weight and fuse modality information. To tackle the issue of insufficient multi-modal tracking data and achieve a robust multi-modal tracker, BAT [13] proposes a bidirectional adapter for multi-modal prompt fusion while keeping the parameters of the pre-trained models frozen. ProFormer [10] utilizes a customized cross-attention mechanism to achieve modal interaction and introduces a dynamically guided learning algorithm to enhance the representation capabilities of each individual modal branch. To reduce the parameter count of multi-modal fusion networks, CMD [24] introduces a compact RGBT tracker and proposes a cross-modal distillation framework to bridge the performance gap between the compact tracker and the powerful two-stream tracker. Despite this, most existing methods overlook the computational resource waste caused by unreliable modality quality in multi-modal tracking and the computational redundancy introduced by deep networks.

B. Early Exiting

Early exiting methods insert decision modules or classifiers in the middle layer of deep networks to exit simple instances that meet the exit criteria, thereby improving inference speed and preserving computing resources. It has been widely applied in the fields of NLP tasks [26]–[30] and CV tasks [31], [32]. For example, ConsistentEE [30] mitigates the gap between training and inference by utilizing a policy network to exit instances at middle layer during training process. Deecap [31] applies early exiting strategy at decoder layer and equips shallow decoder layers with imitation network to simulate deep features. In order to make the exiting more flexible, MuE [32] allows instances to skip the layers in encoder and decoder simultaneously based on input layer-wise similarities. In recent years, early exiting strategy is also applied to SOT for improving model inference speed.

For example, DyTrack [33] inserts a Decisioner after several middle layers to determine whether the instance should exit. BDTrack [34] integrates dynamic early exit modules within each ViT [22] block, excluding the initial n_{enf} blocks, to facilitate early exits from the network when the exit criteria are met. Additionally, it proposes a motion blur-robust ViT to obtain robust feature representation capabilities. Although successful in single-modal object tracking tasks, the potential of early exiting for multi-modal object tracking tasks has yet to be explored. In this paper, we propose a novel dynamic early exiting method for RGBT object tracking tasks.

III. METHODOLOGY

In this section, we initially present an overview of our proposed framework. Subsequently, we present a detailed introduction of the proposed modality exit module and pseudo-label training strategy. Lastly, we introduce the final training objective.

A. Overview

The overall architecture of DRNet is depicted in Fig. 3. It employs an L -layer ViT [22] encoder as the backbone and a fully convolutional network as tracking head to output the final results. We place the proposed modality exit (ME) module at the beginning of the backbone network to discard potential redundant modal information in advance. During the training phase, to maintain consistency between the distributions of training and inference data, we employ early modality exit based on the output of the ME module. Specifically, if the ME module decides to drop a particular modality, the forward propagation of that modality will be halted, and it will not participate in the backward propagation of gradients. As in the training phase, during the inference phase, the model adjusts the utilization of modalities based on the output of the ME module.

B. Modality Exit Module

In order to alleviate the problem of resource waste caused by interfering modality as well as redundant modality in certain scenarios, we propose a Modality Exit (ME) module to determine whether to discard information from one modality, thereby saving computational resources and improving model inference speed.

As is depicted in Fig. 3, the ME module consists of a shared ViT block, a shared linear layer, and a linear projection layer. Due to the lack of discriminative information in the output features of the encoding layer, training the ME module becomes challenging, as the ME module is unaware of the identity of the target being tracked and struggles to discern the characteristics of the background environment in the current frame. Therefore, we utilize a shared ViT block to perform preliminary feature extraction and relation modeling on the output of the encoding layer. Subsequently, the shared linear layer serves to convert modality token embeddings (i.e. x_{rgb}^0 and x_{tir}^0) into token reliability scores. Finally, we employ an additional linear layer to map the token reliability scores of

the two modalities into a classification label, which is used to determine the modal exit conditions. Through experimentation, setting the number of ViT blocks to 1 can achieve a balance between performance and efficiency. Mathematically, this process can be represented as follows:

$$S_{rgb} = \mathfrak{L}_s(\text{ViT}_s(x_{rgb})), \quad (1)$$

$$S_{tir} = \mathfrak{L}_s(\text{ViT}_s(x_{tir})), \quad (2)$$

$$P_{label} = \mathfrak{L}_p([S_{rgb}, S_{tir}]), \quad (3)$$

where $\mathfrak{L}_s(\cdot)$, $\mathfrak{L}_p(\cdot)$, $\text{ViT}_s(\cdot)$, and $[\cdot]$ represent shared linear layer, additional linear projection layer, shared ViT block, and concatenation operation, respectively. And S_{rgb} , S_{tir} , and P_{label} denote reliability scores of RGB tokens, reliability scores of TIR tokens, and the final classification label. It is noteworthy that to enhance module utilization and reduce the waste of computational resources, we propose a **Layer Reuse technique**, where the ViT output from the ME module is utilized in the subsequent backbone tracking network.

After obtaining the final classification label, the utilization of modalities can be determined according to the following rules:

$$use \begin{cases} \text{only RGB,} & MI(P_{label}) = 0 \\ \text{only TIR,} & MI(P_{label}) = 1 \\ \text{both RGB and TIR,} & MI(P_{label}) = 2 \end{cases} \quad (4)$$

where the $MI(\cdot)$ denotes the function that retrieves the indices of the maximum value within a tensor. And the structure of the network can also be adjusted to either a single-modal or multi-modal structure based on the classification label P_{label} . Let ViT_l represent the ViT block at layer l , this process can be expressed as:

$$x_{rgb}^L = \text{ViT}_L \circ \text{ViT}_{L-1} \circ \cdots \circ \text{ViT}_2(x_{rgb}^1) \quad (5)$$

$$x_{tir}^L = \text{ViT}_L \circ \text{ViT}_{L-1} \circ \cdots \circ \text{ViT}_2(x_{tir}^1) \quad (6)$$

$$Output = \begin{cases} \mathcal{H}(x_{rgb}^L), & MI(P_{label}) = 0 \\ \mathcal{H}(x_{tir}^L), & MI(P_{label}) = 1 \\ \mathcal{H}(x_{rgb}^L + x_{tir}^L), & MI(P_{label}) = 2 \end{cases} \quad (7)$$

where the $\mathcal{H}(\cdot)$ denotes the tracking head (Section III-D), and the $\circ$ represents the composition operation. Note that the Eq. 5 and 6 are executed only if the corresponding modality does not undergo early exiting.

C. Pseudo-label Generation

Drawing inspiration from QAT [6], we employ a pseudo-label training approach to stabilize the training process of the ME module. Specifically, to generate pseudo-labels, we first utilize a single-modal ViT tracker loaded with DropTrack [35] weights to directly test on the entire LasHeR [15] training set, obtaining the Intersection over Union (IoU) of bounding boxes for each frame in each modality. Subsequently, we compute the difference in IoU between the RGB modality and the TIR modality for each frame and sort these differences in descending order. Finally, we classify the top 1/3 of frames

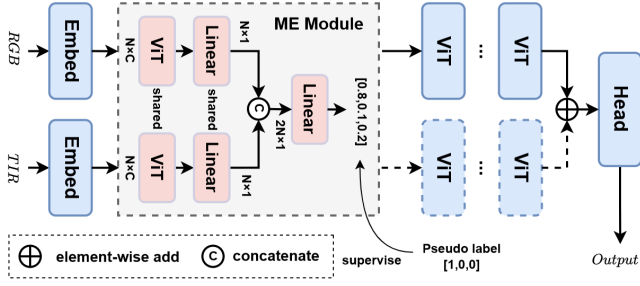


Fig. 3. The detailed architecture of our proposed DRNet model. To ensure the discriminative power of the modal exit module, we employ a single layer of ViT to roughly extract features from the images. The output of this ViT layer will be further utilized for subsequent fine-grained feature extraction, thereby reducing the waste of computational resources.

as *rgb-only* (indicating that the RGB modality is more reliable and only RGB modal data is used), the bottom 1/3 as *tir-only*, and the remaining frames as *rgb-tir* (indicating that both modalities are reliable). The above generation algorithm is illustrated in Algorithm 1, where GAP_{iou} and GAP_{new} are Python dictionaries, the keys record the information of sequences and frames, and the values record the IoU difference. Subsequently, we utilize the obtained pseudo-labels to supervise the output of the ME module. The objective loss is formulated as:

$$\mathcal{L}_{ME} = CE(P_{label}, y). \quad (8)$$

Here, $CE(\cdot)$ is the cross-entropy loss, P_{label} is the output of the ME module, and y is the generated pseudo-label.

Algorithm 1 Pseudo-label Generation of DRNet-Base.

Input: IoU of each frame for two modalities

Output: The expected pseudo-labels for modality usage, i.e.

```

 $GAP_{new}$ 
1:  $GAP_{new} \leftarrow$  empty dictionary;
2:  $GAP_{iou} \leftarrow IoU_{rgb} - IoU_{tir}$ ;
3:  $List_{desc} \leftarrow$  Sort  $GAP_{iou}$  in descending order by key (i.e. IoU gap);
4: for  $i \leftarrow 1$  to  $\lfloor \text{len}(List_{desc})/3 \rfloor$  do
5:    $rgb_{only} \leftarrow List_{desc}[i]$ ;
6:    $tir_{only} \leftarrow List_{desc}[-(i+1)]$ ;
7:   if  $rgb_{only} > 0$  then
8:      $GAP_{new}[rgb_{only}[0]] \leftarrow 0$ ; // only use RGB
9:   end if
10:  if  $rgb_{tir} < 0$  then
11:     $GAP_{new}[tir_{only}[0]] \leftarrow 1$ ; // only use TIR
12:  end if
13: end for
14: for  $j \leftarrow 1$  to  $\text{len}(GAP_{desc\_list})$  do
15:  if  $GAP_{new}$  doesn't have key  $List_{desc}[j][0]$  then
16:     $GAP_{new}[List_{desc}[j][0]] \leftarrow 2$ ; // use both RGB and TIR
17:  end if
18: end for
19: return  $GAP_{new}$ 

```

However, although Algorithm 1 addresses the modality interference issue mentioned in Section 1, it can not resolve the

problem of modality redundancy. Therefore, we further modify the algorithm (please refer to Algorithm 2, where the text in red indicates newly added modifications) by categorizing frames with a modality IoU difference of 0 into the *rgb-only* class. This is based on empirical evidence that when the modal quality difference is minimal, RGB texture information is generally considered more abundant. For ease of distinction, we refer to the model trained using pseudo-labels generated by Algorithm 1 as **DRNet-Base**, and the model trained using pseudo-labels generated by Algorithm 2 as **DRNet-Fast**.

Algorithm 2 Pseudo-label Generation of DRNet-Fast.

Input: IoU of each frame for two modalities

Output: The expected pseudo-labels for modality usage, i.e.

```

 $GAP_{new}$ 
1:  $GAP_{new} \leftarrow$  empty dictionary;
2:  $GAP_{iou} \leftarrow IoU_{rgb} - IoU_{tir}$ ;
3:  $List_{desc} \leftarrow$  Sort  $GAP_{iou}$  in descending order by key (i.e. IoU gap);
4: for  $i \leftarrow 1$  to  $\text{len}(List_{desc})$  do
5:   if  $i < \lfloor \text{len}(List_{desc})/3 \rfloor$  then
6:      $rgb_{only} \leftarrow List_{desc}[i]$ ;
7:      $tir_{only} \leftarrow List_{desc}[-(i+1)]$ ;
8:     if  $rgb_{only} > 0$  then
9:        $GAP_{new}[rgb_{only}[0]] \leftarrow 0$ ; // only use RGB
10:    end if
11:    if  $rgb_{tir} < 0$  then
12:       $GAP_{new}[tir_{only}[0]] \leftarrow 1$ ; // only use TIR
13:    end if
14:  else
15:     $item \leftarrow List_{desc}[i]$ ;
16:    if  $item[1] == 0$  then
17:       $GAP_{new}[item[0]] \leftarrow 0$ ; // only use RGB
18:    end if
19:  end if
20: end for
21: for  $j \leftarrow 1$  to  $\text{len}(GAP_{desc\_list})$  do
22:  if  $GAP_{new}$  doesn't have key  $List_{desc}[j][0]$  then
23:     $GAP_{new}[List_{desc}[j][0]] \leftarrow 2$ ; // use both RGB and TIR
24:  end if
25: end for
26: return  $GAP_{new}$ 

```

D. Prediction Head and Loss Function

The features of the backbone network are sent to a tracking head, which consists of three fully convolutional neural network branches and outputs a classification score map, normalized bounding box size, and offset size. Finally, based on these outputs, the bounding box is calculated. Similar to [36], we utilize the weighted focal loss [37] for classification, and the L1 loss and generalized IoU loss [38] for regression. The overall loss of the training process is:

$$\mathcal{L}_{task} = \mathcal{L}_{cls} + \lambda_{iou}\mathcal{L}_{iou} + \lambda_{L1}\mathcal{L}_1. \quad (9)$$

$$\mathcal{L}_{total} = \mathcal{L}_{ME} + \mathcal{L}_{task}. \quad (10)$$

TABLE I

COMPARISON WITH TRADITIONAL STATE-OF-THE-ART METHODS ON RGBT210, RGBT234, AND LASHER DATASETS. $\uparrow$ REPRESENTS THE LARGER THE BETTER AND THE TOP TWO RESULTS ARE MARKED IN **Red** AND **Blue** FONTS. * DENOTES THAT WE UTILIZE THE MODEL PROVIDED BY THE OFFICIAL SOURCE TO CONDUCT RE-TESTING IN OUR ENVIRONMENT, AND THE PARENTHESES () INDICATE THAT THE METHOD IN THE PAPER ONLY PRESENTS AN OVERALL FPS DATA WITHOUT SPECIFYING WHICH DATASET THE RESULT IS BASED ON.

Methods	Source	Backbone	RGBT210			RGBT234			LasHeR			
			PR $\uparrow$	SR $\uparrow$	FPS $\uparrow$	MPR $\uparrow$	MSR $\uparrow$	FPS $\uparrow$	PR $\uparrow$	NPR $\uparrow$	SR $\uparrow$	FPS $\uparrow$
CAT [16]	ECCV 2020	VGG-M	79.2	53.3	-	80.4	56.1	-	-	-	-	-
MANet++ [17]	TIP 2021	VGG-M	-	-	-	80.0	55.4	-	46.7	40.4	31.4	(25.4)
APFNet [18]	AAAI 2022	VGG-M	-	-	-	82.7	57.9	-	50.0	-	36.2	-
ProTrack [39]	ACM MM 2022	ViT-Base	-	-	-	78.6	58.7	-	50.9	-	42.1	-
HMFT [19]	CVPR 2022	ResNet-50	78.6	53.5	-	78.8	56.8	-	-	-	-	(30.2)
MACFT [40]	Sensors 2023	ViT-Base	-	-	-	85.7	62.2	-	65.3	-	51.4	33.3
CMD [24]	CVPR 2023	ResNet-50	-	-	-	82.4	58.4	-	59.0	54.6	46.45	(30.0)
ViPT [20]	CVPR 2023	ViT-Base	79.8*	57.9*	34.4*	83.5	61.7	34.5*	65.1	61.6*	52.5	31.6*
TBSI [14]	CVPR 2023	ViT-Base	85.3	62.5	31.0*	87.1	63.7	30.9*	69.2	65.7	55.6	30.8*
BAT [13]	AAAI 2024	ViT-Base	84.5*	61.0*	33.7*	86.8	64.1	34.0*	70.2	-	56.3	32.5*
SDSTrack [21]	CVPR 2024	ViT-Base	81.3	58.4	32.9*	84.8	62.5	32.6*	66.5	-	53.1	31.6*
Un-Track [41]	CVPR 2024	ViT-Base	-	-	-	84.2	62.5	-	66.7	-	53.6	-
OneTracker [25]	CVPR 2024	ViT-Base	-	-	-	85.7	64.2	-	67.2	-	53.8	-
DRNet-Fast	Ours	ViT-Base	82.4	59.0	74.4	85.9	63.3	75.6	67.1	62.8	53.8	70.9
DRNet-Base		ViT-Base	85.0	60.0	67.9	87.7	64.5	66.9	69.2	65.0	55.6	62.2

where λ_{iou} and λ_{L_1} are hyper-parameters, which we set to 2 and 5, respectively.

IV. EXPERIMENTS

A. Implementation Details

We implement our model on two NVIDIA GeForce RTX 4090 GPUs. We initialize DRNet using the DropTrack [35] pre-trained model and train it on the LasHeR training set, with the learning rate set to $1e-5$ for the backbone while $1e-4$ is used for others, and the AdamW [42] optimizer employed. When fine-tuning the Baseline (for comparative experiments), we train for 15 epochs with a batch size of 15, and reduce the learning rate by 10 times after 12 epochs. During the training phrase of DRNet, we set the batch size to 32 for 50 epochs, and the learning rate is decreased by 10 times after 40 epochs. The aforementioned batch sizes are solely chosen to accommodate the GPU memory size, without any specific design considerations. During the inference stage, to obtain stable FPS results, we opt to test on an idle NVIDIA TITAN RTX GPU.

B. Datasets and Evaluation Metrics

We conduct extensive experiments on three standard RGBT tracking datasets, namely RGBT210 [43], RGBT234 [44], and LasHeR [15], to validate the effectiveness of our method. In addition, we utilize six metrics to evaluate the model's performance, including Precision Rate (PR), Success Rate (SR), Normalized Precision Rate (NPR), Maximum Precision Rate (MPR), Maximum Success Rate (MSR), and Frames Per Second (FPS).

1) *RGBT210*: RGBT210 dataset contains 210 RGBT video sequences with more than 209K frames, annotated with 12 attributes. This dataset uses two evaluation metrics, namely PR and SR. **PR** evaluates the proportion of frames where the distance between the center of the bounding box predicted

by the model and the ground truth is lower than a specified threshold. **SR** measures the percentage of successfully tracked frames whose overlap between the output bounding box and the ground truth is greater than a certain threshold.

2) *RGBT234*: RGBT234 comprises 234 RGBT video sequence pairs, along with their corresponding ground truth annotations and 12 challenge attributes. The total number of frames amounts to 234K, with the longest video sequence containing 8K frames. It employs **MPR** and **MSR** as its evaluation criteria. More precisely, from the RGB and Thermal modalities, it selects the higher PR and SR as the definitive metrics.

3) *LasHeR*: LasHeR is a large-scale RGBT tracking dataset that contains 1,224 RGBT video sequence pairs, totaling approximately 730K frames. It is divided into 245 test sequences and 979 training sequences. Compared to the previous work [44], it offers seven newly labeled challenge attributes: abrupt illumination variation (AIV), aspect ratio change (ARC), frame lost (FL), high illumination (HI), hyaline occlusion (HO), out-of-view (OV), and similar appearance (SA). For evaluation, it employs **PR**, **SR**, and **NPR**, where NPR is the normalized version of PR.

C. Comparison with Traditional Methods

We compare our method with previous traditional methods on three RGBT tracking datasets, and the comparison results are presented in Table I. Depending on the different pseudo-label generation algorithms employed, we denote the model trained using Algorithm 1 as DRNet-Base and the model trained using Algorithm 2 as DRNet-Fast. It is noteworthy that, in order to mitigate the impact of device randomness, the FPS of our method and all other re-tested FPS values are calculated as the average FPS across all sequences in the corresponding test set.

1) *Quantitative Evaluation on RGBT210*: As shown in Table I, compared to the recent method BAT, our DRNet-Base achieves only 1.0% lower in the SR metric, while outperforming it by 0.5% in the PR metric, and boasts a 2.0x faster inference speed. When compared to TBSI, which is the top-performing algorithm in this dataset, our DRNet-Base exhibits only a 0.3%/2.5% decrease in PR/SR, yet it achieves a speed that is 2.2 times that of TBSI. Even DRNet-Fast, which only uses single modality in most video frames (see Fig. ??), exhibits only a 2.1% decrease in the PR and a 2.0% decrease in the SR compared to BAT, while achieving an improvement of 40.7 FPS in speed. Additionally, Compared to SDSTrack, our DRNet-Fast comprehensively surpasses it across all metrics, achieving a $2.3\times$ faster inference speed while improving by 1.4% and 0.6% in PR and SR, respectively. This fully demonstrates the effectiveness of our method in reducing computational overhead while maintaining performance.

2) *Quantitative Evaluation on RGBT234*: On the RGBT234 dataset, compared to the methods listed in the table, our DRNet-Base not only achieves a speed that is 2.2 times faster than TBSI but also ranks first in performance. This further demonstrates the exceptional performance of our method in balancing accuracy and speed. Furthermore, our DRNet-Fast offers a speed that is 2.2 times faster than BAT, while only exhibiting a performance decrease of 0.9% MPR and 0.8% MSR compared to BAT. When compared to SDSTrack and Un-Track, our DRNet-Fast outperforms both in terms of MPR and MSR, while achieving a speed of 75.6 FPS, which is 2.3 times faster than SDSTrack.

3) *Quantitative Evaluation on LasHeR*: LasHeR is a challenging and high-quality dataset, resulting in a decrease in early-exited instances compared to the previous two datasets, as shown in Fig. 4. Despite this, our DRNet-Base remains comparable in performance to TBSI while achieving a speed that is twice as fast. Even when compared to the highest-performing method, BAT, DRNet-Base only falls short by 1.0% in PR and 0.7% in SR, yet its speed reaches 1.9 times that of BAT. For DRNet-Fast, it outperforms MACFT by 1.8% in PR and 2.4% in SR, while still maintaining a speed that is 2.1 times faster. In addition, the speed of DRNet-Fast is 2.2 times that of ViPT and SDSTrack. Meanwhile, DRNet-Fast outperforms ViPT by 2.0%, 1.2%, and 1.3% on PR/NPR/SR, respectively, and surpasses SDSTrack by 0.6% and 0.7% on PR and SR, respectively. This demonstrates the versatility and adaptability of our method across various RGBT datasets.

4) *Efficiency Comparison*: To highlight the superiority of our method in conserving computational resources, we compare the FLOPs and parameter counts of our method with four recent state-of-the-art methods (i.e. ViPT [20], TBSI [14], BAT [13], and SDSTrack [21]) based on ViT-Base [22] in Table II. Due to the capability of our method to dynamically adjust the network structure based on input quality, its FLOPs and parameter count vary within a specific range of values. Our approach demonstrates lower FLOPs and parameter counts compared to both TBSI and BAT. Notably, when compared to TBSI, our DRNet achieves comparable performance yet with only 0.46 times its parameter count, while also exhibiting clear advantages in terms of FLOPs and speed. Regarding

TABLE II
COMPARISON OF FLOPs AND PARAMETERS ON NVIDIA GeForce RTX 4090 GPU.

Methods	Source	Backbone	FLOPs↓	Params.↓
ViPT	CVPR2023	ViT-Base	21.8G	93.0M
TBSI	CVPR2023	ViT-Base	82.5G	202.0M
BAT	AAAI2024	ViT-Base	56.7G	92.4M
SDSTrack	CVPR2024	ViT-Base	54.4G	102.2M
Baseline		ViT-Base	56.4G	92.1M
DRNet-Base	Ours	ViT-Base	31.5G or 56.4G	92.1M
DRNet-Fast		ViT-Base	31.5G or 56.4G	92.1M

SDSTrack, although its FLOPs are lower than the maximum FLOPs of our method, our approach has the capability to adaptively adjust the model structure based on the input, thereby maintaining relatively low FLOPs in most cases. Furthermore, we possess a notable advantage in terms of speed. Since the auxiliary modality in ViPT is not directly processed through the computationally intensive ViT network, but instead are handled using a few simple convolutional layers and then summed with RGB modality features before being input into the ViT encoder, its FLOPs are relatively low. However, our DRNet boasts fewer parameters and a significant speed advantage. Finally, compared to the baseline, our DRNet has a comparable number of parameters and maximum FLOPs, indicating that the introduction of our ME module introduces negligible additional parameters to the network. However, it significantly enhances the inference speed of the network and reduces the average computational load.

D. Ablation Study

As shown in Table III, taking DRNet-Base as an example, we conduct ablation studies on three datasets to evaluate the importance of ME module and ViT block in ME module.

Baseline refers to the utilization of a shared ViT-Base backbone for extracting features from two modalities, with the features of the two modalities being summed before being fed into the tracking head, without any modal interaction or early exit.

w/ME-raw denotes the utilization of the ME module without incorporating ViT, meaning that no ViT blocks are employed for preliminary feature extraction and relationship modeling. This results in insufficient discriminative power in the input features of the ME-raw module, which can adversely affect the training effectiveness and even lead to ineffective training (as calculated, the ME_raw module classifies almost all frames indiscriminately as *rgb_only*). Therefore, despite its fast inference speed, there is a significant decrease in tracking accuracy and success rate, and it fails to achieve network adaptability to different scenarios.

w/ME (DRNet-Base) represents the baseline network equipped with a complete ME module. It is observed that compared to the *Baseline*, the incorporation of the ME module significantly enhances the inference speed while maintaining a high tracking accuracy. For example, on the RGBT234 dataset, compared to the baseline, DRNet-Base achieves an increase of nearly 20 FPS in speed with only a 0.7% and 1.6%

TABLE III
ABLATION STUDY ON THE ME MODULE OF THE DRNet-BASE MODEL ON THREE DATASETS. THE ROW WITH A GREY BACKGROUND REPRESENTS OUR DRNet-BASE MODEL.

Versions	ME-raw	ME	RGBT210			RGBT234			LasHeR			
			PR↑	SR↑	FPS↑	MPR↑	MSR↑	FPS↑	PR↑	NPR↑	SR↑	FPS↑
Baseline	-	-	87.1	63.4	46.6	88.4	66.1	46.6	71.2	67.4	57.1	46.7
w/ME-raw	✓	-	79.9	58.5	79.2	83.1	61.5	79.8	65.8	61.4	52.6	77.1
w/ME	-	✓	85.0	60.0	67.9	87.7	64.5	66.9	69.2	65.0	55.6	62.2

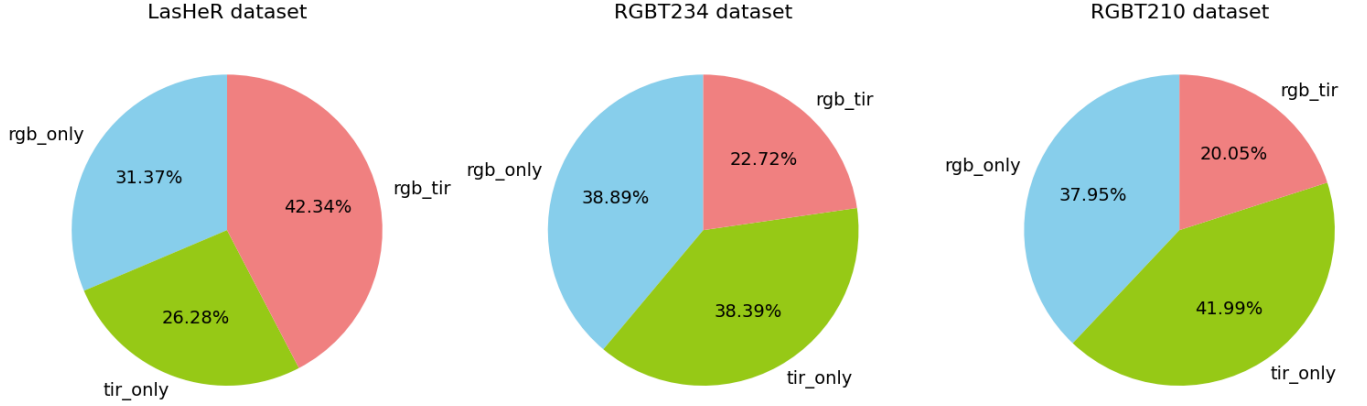


Fig. 4. Visualization of modality usage across three RGBT datasets for the DRNet-BASE model. The bright coral-colored sections indicate no early exit of modalities, the light green sections represent the usage of only the TIR modality, and the sky blue sections signify the usage of only the RGB modality.

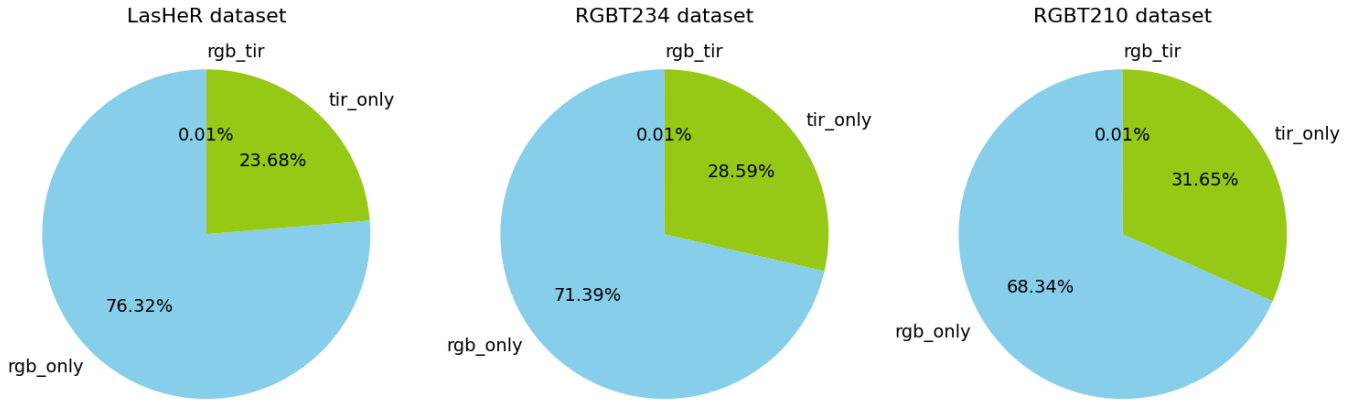


Fig. 5. Visualization of modality usage across three RGBT datasets for the DRNet-Fast model.

decrease in MPR and MSR, respectively. This fully validates the effectiveness of our proposed modality exit module in maintaining tracking accuracy, enhancing inference speed, and conserving computational resources. Furthermore, DRNet-Base exhibits significant performance advantages compared to *w/ME-raw*. For instance, on the LasHeR dataset, DRNet-Base surpasses *w/ME-raw* by 3.4%/3.6%/3.0% in terms of the PR/NPR/SR metrics. This superiority is attributed to the capability of the ME module to dynamically utilize modality information according to the scene context.

E. Analysis and Visualization

1) *Model Speed Visualization*: To more intuitively observe the effectiveness of our method in balancing speed and accuracy, we have visualized the PR-FPS plots for our method along with numerous state-of-the-art algorithms. As illustrated in Fig. 2, our model DRNet-Base is capable of maintaining performance comparable to TBSI [14] while achieving a speed more than twice as fast. Moreover, the DRNet-Fast exhibits accuracy similar to SDSTrack [21] but operates at a speed more than 2.2 times faster.

2) *Modality Usage Visualization*: To better observe the modal exit ratios of the model across different datasets, we

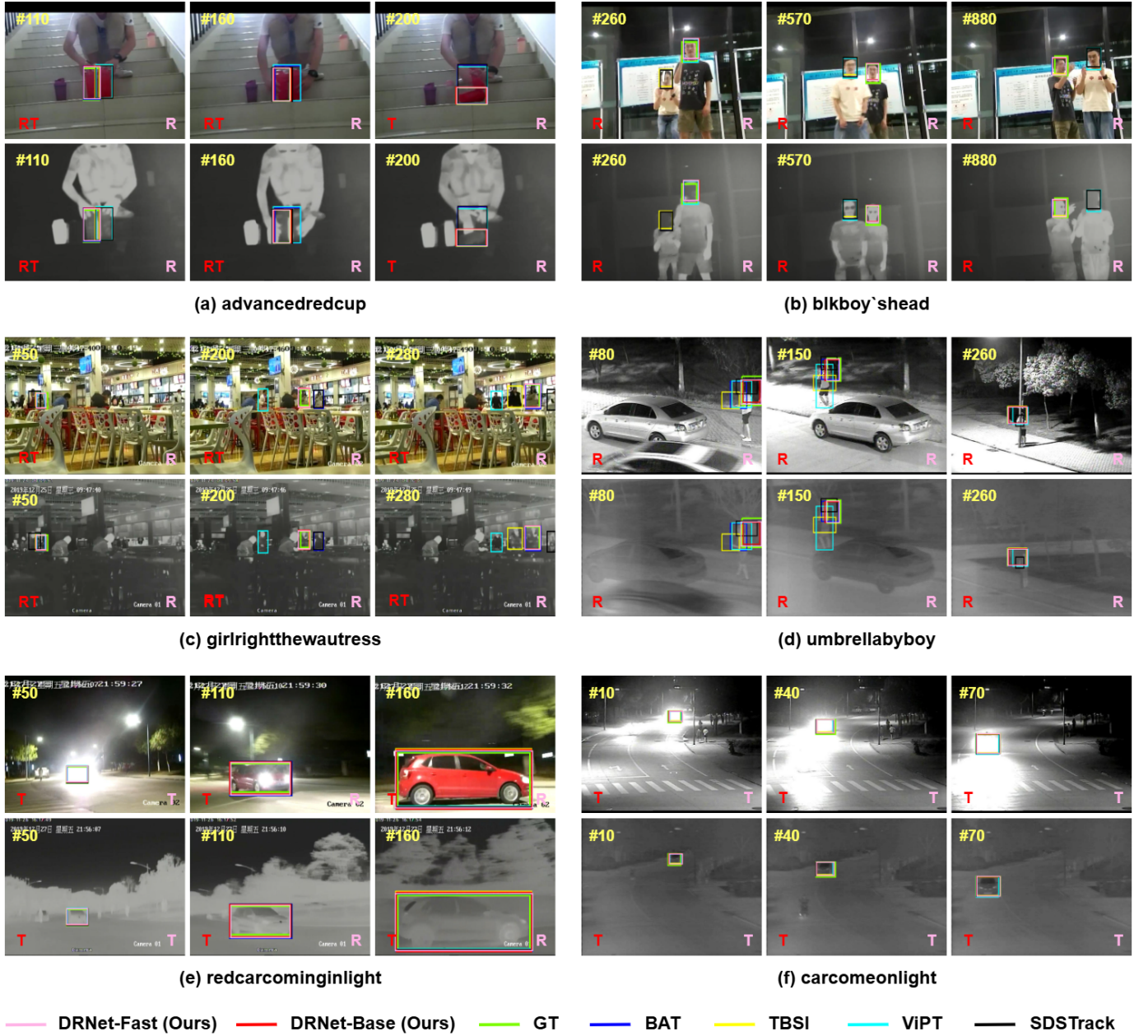


Fig. 6. Qualitative comparison between our method and other RGBT trackers on four representative sequences from LasHeR dataset. The green bounding boxes denote ground truth. In each frame of the image, the red marker in the lower left corner indicates the modality usage of DRNet-Base, while the pink marker in the lower right corner represents the modality usage of DRNet-Fast. Specifically, RT, R, and T denote the utilization of both RGB and TIR modalities, the exclusive use of the RGB modality, and the exclusive use of the TIR modality, respectively.

visualize the modal usage of DRNet-Base and DRNet-Fast on three datasets, as shown in Fig. 4 and 5. In Fig. 4, DRNet-Base trained using Algorithm 1 exhibits modal early exit in over half of the test frames, which explains the significant speed-up achieved by our model. However, on the LasHeR dataset, due to its greater challenge compared to the other two datasets, modal dropout does not occur in 42.34% of the frames. To address this, we have improved the pseudo-label generation algorithm. The DRNet-Fast model trained using Algorithm 2 achieves modal dropout in almost all frames, as shown in Fig. 55. This enables DRNet Fast to further enhance the inference speed by approximately 14% on the LasHeR

dataset, and by roughly 13% and 10% on the RGBT234 and RGBT210 datasets, respectively.

3) *Qualitative Comparison*: Even though our model predominantly utilizes information from a single modality for tracking in most cases, it still demonstrates robust tracking performance and exhibits strong scene adaptability. As shown in Fig. 6(a), the scene presents challenges due to similar object interference, and the quality difference between the two modalities is insignificant. Therefore, DRNet-Base, trained based on Algorithm 1, utilizes information from both RGB and TIR modalities in most frames. In contrast, DRNet-Fast solely relies on RGB modality information, as the quality

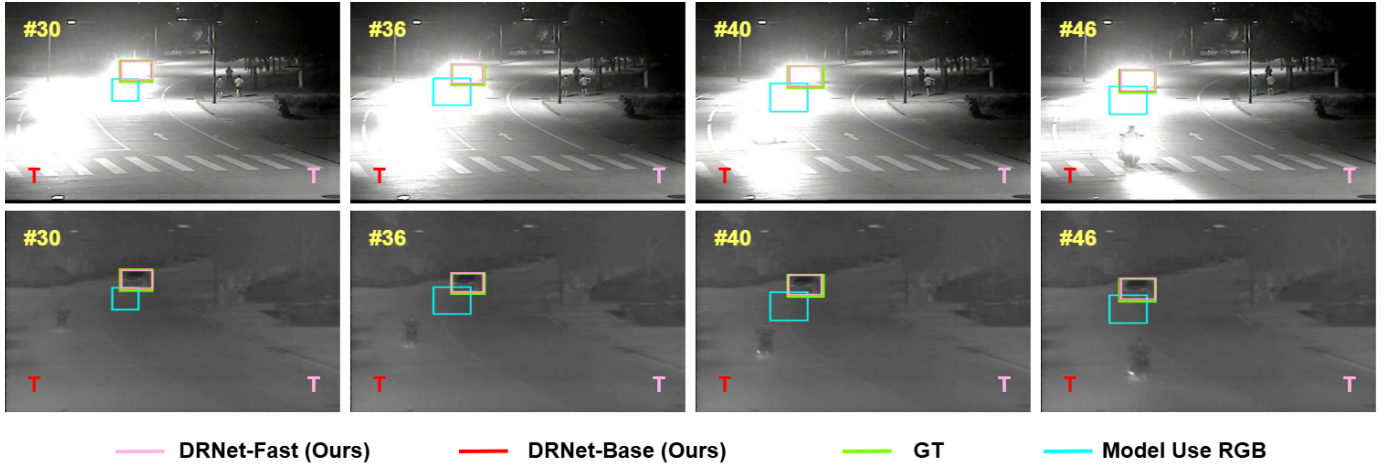


Fig. 7. Visualization of tracking results for DRNet-Base model, DRNet-Fast model, and model using only RGB modality.

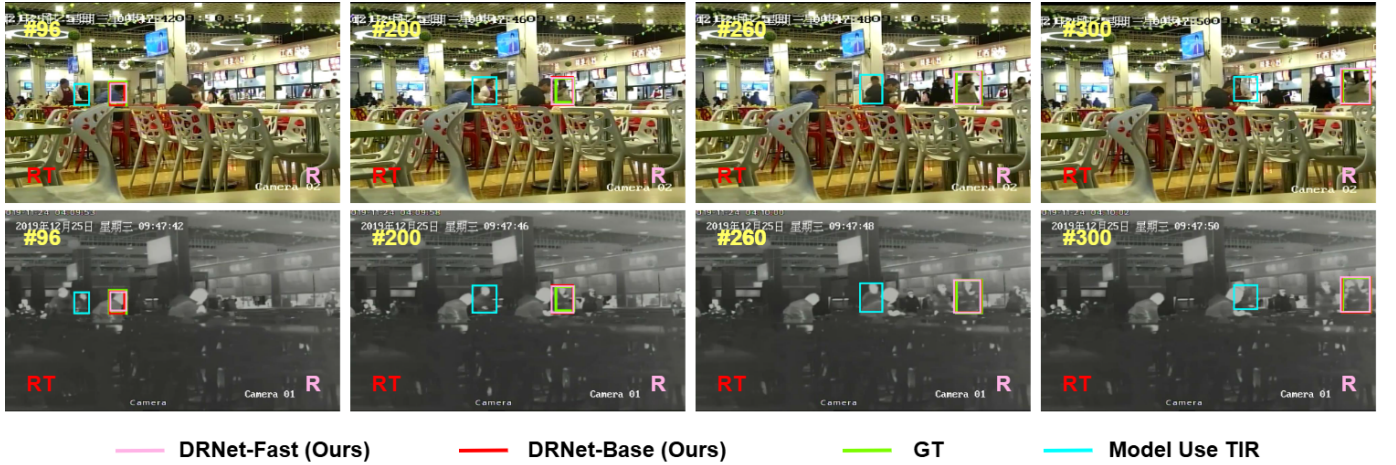


Fig. 8. Visualization of tracking results for DRNet-Base model, DRNet-Fast model, and model using only TIR modality.

difference between the two modalities is minimal, prompting modal to classify them as *rgb_only*. In Fig. 6(b), the presence of glass occlusion and similar object interference results in significant loss of information in the TIR modality, leading the model to select the RGB modality with richer texture information for tracking. In Fig. 6(c), DRNet-Base opts to utilize information from both modalities. In contrast, DRNet-Fast selects the RGB modality only, which is richer in texture information, in order to reduce computational load. In Fig. 6(d), since the tracked umbrella is not a heat source, the TIR modality is nearly unusable, prompting the model to choose the RGB modality as input. In Fig. 6(e), when the RGB images in previous frames fail to provide effective target information, the model selects the TIR modality. After frame 110, using the RGB modality becomes more effective. In Fig. 6(f), the RGB modality fails to provide valid target information throughout, leading the model to use only the TIR modality for tracking. These examples fully demonstrate the tracking robustness and high scene adaptability of our method.

4) *Scene Adaptive Visualization*: To demonstrate the adaptive capability of our DRNet to various scenarios, we trained

two single-modal models on the LasHeR training set: one exclusively using RGB data and the other exclusively using TIR data. Subsequently, we visualized their tracking results on the LasHeR testing set, as illustrated in Fig. 7 and 8. In the scenario illustrated in Fig. 7, the single-modal model struggles to locate the target using only RGB modality information, ultimately leading to tracking failure. In contrast, our model adaptively switches to utilizing TIR modality information, thereby achieving stable tracking. In the scenario depicted in Fig. 8, it is difficult to discern the target in both modality images. The DRNet-Base model considers the quality of the two modalities to be comparable in these frames and categorizes them into the *rgb_tir* class. In contrast, DRNet-Fast categorizes them as *rgb_only*, due to the RGB modality possessing more texture information. However, the single-modal model solely utilizing the TIR modality fails to robustly track the target in this scenario, highlighting the importance of model adaptability to different scenarios.

V. CONCLUSION

In this work, we propose a novel adaptive multi-modal early exit mechanism for RGBT tracking. Most existing methods

overlook the issues of computational waste caused by unreliable modalities and computational redundancy arising from modalities of equivalent quality. To address these problems, we introduce a modality exit module to adaptively terminate the forward propagation process of these modality data in advance. Furthermore, we propose a layer reuse technique to save computational costs and achieve a balance between speed and accuracy. Additionally, we present a pseudo-label training strategy to enhance the robustness of the modality exit module and stabilize the training process. Extensive experiments validate that our method can significantly improve inference speed while maintaining acceptable accuracy levels. We believe that our DRNet can be easily transferred to other multi-modal tracking frameworks. In future work, we plan to explore superior implementation schemes to further enhance inference speed and tracking accuracy, and to verify the generality of our method across various multi-modal tracking datasets.

ACKNOWLEDGMENTS

The authors would like to thank the anonymous reviewers for their contributions to improving the quality of this article.

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